

Step-by-Step Guide: Creating Google Calendar to Salesforce Event Workflow in n8n

This guide walks you through creating the complete workflow in n8n from scratch.

Prerequisites Checklist

Before starting, ensure you have:

- ☐ Access to n8n instance (cloud or self-hosted)
 - ☐ Google Calendar account with API access
 - ☐ Salesforce org with API access
 - ☐ Slack workspace (for notifications)
 - ☐ Admin access to create custom fields in Salesforce (optional)
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Part 1: Salesforce Setup (Do This First)

Step 1.1: Create Custom Fields in Salesforce

1. Log into Salesforce

- Go to Setup (gear icon → Setup)

2. Navigate to Object Manager

- In Quick Find, type "Object Manager"
- Click on "Object Manager"

3. Select Event Object

- Find and click on "Event" in the list

4. Create First Custom Field: Google Calendar Event ID

- Click "Fields & Relationships" in the left sidebar
- Click "New" button
- Select "Text" as the field type
- Click "Next"
- **Field Label:** Google Calendar Event ID
- **Field Name:** Google_Calendar_Event_ID (auto-populated)
- **Length:** 255
- **Required:** Unchecked
- Click "Next"
- **Field-Level Security:** Check "Visible" for profiles that need access
- Click "Next"
- **Page Layout Assignment:** Add to relevant page layouts
- Click "Save"

5. Create Second Custom Field: Google Calendar Event Link

- Click "New" again
- Select "URL" as the field type
- Click "Next"
- **Field Label:** Google Calendar Event Link
- **Field Name:** Google_Calendar_Event_Link (auto-populated)
- **Required:** Unchecked
- Click "Next"
- **Field-Level Security:** Check "Visible" for profiles that need access
- Click "Next"
- **Page Layout Assignment:** Add to relevant page layouts
- Click "Save"

6. Note the Field API Names

- Google_Calendar_Event_ID__c
- Google_Calendar_Event_Link__c
- You'll need these in the workflow

Part 2: n8n Workflow Creation

Step 2.1: Create New Workflow

1. Open n8n

- Navigate to your n8n instance
- Log in if needed

2. Create New Workflow

- Click "Workflows" in the left sidebar
- Click the "+" button or "Add Workflow"
- Name it: Google Calendar to Salesforce Event

3. Save the Workflow

- Click "Save" (top right) to save the empty workflow

Part 2.2: Set Up Google Calendar OAuth2 Credentials

1. Go to Credentials

- Click the menu icon (≡) in the top left
- Select "Credentials"

2. Add Google Calendar Credential

- Click "Add Credential"
- Search for "Google Calendar"

- Select "Google Calendar OAuth2 API"

3. Configure OAuth2

- **Credential Name:** Google Calendar OAuth2 API
- Click "Connect my account"
- You'll be redirected to Google sign-in
- Sign in with the Google account that has the calendar you want to monitor
- Grant permissions:
 - ☒ View your calendars
 - ☒ View events on your calendars
- Click "Allow"
- You'll be redirected back to n8n
- The credential should now be saved

4. Test the Connection

- The credential should show as "Connected"
 - If there's an error, check your Google account permissions
-

Part 2.3: Set Up Salesforce OAuth2 Credentials

1. Add Salesforce Credential

- Still in Credentials section
- Click "Add Credential"
- Search for "Salesforce"
- Select "Salesforce OAuth2 API"

2. Configure Salesforce OAuth2

- **Credential Name:** Salesforce OAuth2 API
- **Environment:** Select "Production" or "Sandbox" based on your org
- Click "Connect my account"
- You'll be redirected to Salesforce login
- Log in with your Salesforce credentials
- Click "Allow" to grant access
- You'll be redirected back to n8n
- The credential should now be saved

3. Test the Connection

- The credential should show as "Connected"
 - If there's an error, verify your Salesforce org URL and permissions
-

Part 2.4: Set Up Slack Credentials (Optional but Recommended)

1. Add Slack Credential

- Still in Credentials section
- Click "Add Credential"
- Search for "Slack"
- Select "Slack API"

2. Configure Slack API

- **Credential Name:** Slack API
- **Authentication:** Select "OAuth2" or "Access Token"

Option A: OAuth2 (Recommended)

- Click "Connect my account"
- Authorize n8n in Slack
- Select the workspace
- Grant permissions: chat:write, chat:write.public
- Click "Allow"

Option B: Access Token

- Create a Slack App at <https://api.slack.com/apps>
- Generate a Bot Token
- Paste the token in the credential

3. Test the Connection

- The credential should show as "Connected"

Part 3: Building the Workflow Nodes

Step 3.1: Add Google Calendar Trigger Node

1. Add Trigger Node

- Go back to your workflow
- Click the "+" button in the canvas
- Search for "Google Calendar"
- Select "Google Calendar Trigger"

2. Configure the Trigger

- **Node Name:** Google Calendar Trigger
- **Credential:** Select "Google Calendar OAuth2 API" (the one you created)
- **Event:** Select "Event Created"
- **Calendar:** Leave as "primary" or select a specific calendar ID
- Click "Execute Node" to test (this will show sample data)

3. Save the Node

- Click outside the node or press Ctrl+S to save

Step 3.2: Add Format Event Data Code Node

1. Add Code Node

- Click the "+" button after the Google Calendar Trigger
- Search for "Code"
- Select "Code"

2. Configure the Code Node

- **Node Name:** `Format Event Data`
- **Mode:** Select "Run Once for All Items" or "Run Once for Each Item"
- **Language:** JavaScript

3. Add the JavaScript Code

- In the code editor, paste the following:

```
// Format Google Calendar event data for Salesforce Event
const googleEvent = $input.item.json;

// Extract relevant data from Google Calendar event
const eventData = {
  Subject: googleEvent.summary || 'Untitled Event',
  Description: googleEvent.description || '',
  StartDateTime: googleEvent.start?.dateTime || googleEvent.start?.date,
  EndDateTime: googleEvent.end?.dateTime || googleEvent.end?.date,
  Location: googleEvent.location || '',
  IsAllDayEvent: !googleEvent.start?.dateTime && !!googleEvent.start?.date,
  // Map Google Calendar attendees to Salesforce Event attendees
  Attendees: googleEvent.attendees?.map(attendee => attendee.email) || [],
  // Store Google Calendar event ID for reference
  GoogleCalendarEventId: googleEvent.id,
  GoogleCalendarEventLink: googleEvent.htmlLink || '',
  // Additional metadata
  OrganizerEmail: googleEvent.organizer?.email || '',
  Status: googleEvent.status || 'confirmed'
};

return {
  json: eventData
};
```

4. Test the Node

- Click "Execute Node" to test with sample data
- Verify the output shows the formatted data structure

5. Save the Node

Step 3.3: Add Create Salesforce Event Node

1. Add Salesforce Node

- Click the "+" button after the Format Event Data node
- Search for "Salesforce"
- Select "Salesforce"

2. Configure the Salesforce Node

- **Node Name:** Create Salesforce Event
- **Credential:** Select "Salesforce OAuth2 API"
- **Resource:** Select "Event"
- **Operation:** Select "Create"

3. Configure Fields

- Click "Add Field" for each field you want to map:

Required Fields:

- **Subject:** Click the field, then click the expression icon (fx), enter: `{{ $json.Subject }}`
- **StartDateTime:** Expression: `{{ $json.StartDateTime }}`
- **EndDateTime:** Expression: `{{ $json.EndDateTime }}`

Optional Fields:

- **Description:** Expression: `{{ $json.Description }}`
- **Location:** Expression: `{{ $json.Location }}`
- **IsAllDayEvent:** Expression: `{{ $json.IsAllDayEvent }}`

Custom Fields:

- **Google_Calendar_Event_ID__c:** Expression: `{{ $json.GoogleCalendarEventId }}`
- **Google_Calendar_Event_Link__c:** Expression: `{{ $json.GoogleCalendarEventLink }}`

4. Alternative: Use Fields JSON

- Instead of adding fields one by one, you can click "Fields (JSON)" tab
- Paste this JSON:

```
{
  "Subject": "={{ $json.Subject }}",
  "Description": "={{ $json.Description }}",
  "StartDateTime": "={{ $json.StartDateTime }}",
  "EndDateTime": "={{ $json.EndDateTime }}",
  "Location": "={{ $json.Location }}",
  "IsAllDayEvent": "={{ $json.IsAllDayEvent }}",
  "Google_Calendar_Event_ID__c": "={{ $json.GoogleCalendarEventId }}",
  "Google_Calendar_Event_Link__c": "={{ $json.GoogleCalendarEventLink }}"
}
```

5. Test the Node (Optional - will create a real event in Salesforce)

- Click "Execute Node" to test
- ⚠ **Warning:** This will create an actual event in Salesforce!
- Check your Salesforce org to verify the event was created

6. Save the Node

Step 3.4: Add Slack Alert Node (Success Notification)

1. Add Slack Node

- Click the "+" button after the Create Salesforce Event node
- Search for "Slack"
- Select "Slack"

2. Configure the Slack Node

- **Node Name:** Slack Alert
- **Credential:** Select "Slack API"
- **Resource:** Select "Message"
- **Operation:** Select "Post"

3. Configure Message

- **Channel:** Enter channel name (e.g., #n8n-alerts) or use expression: `{{ $env.SLACK_CHANNEL || '#n8n-alerts' }}`
- **Text:** Click the expression icon (fx) and enter:

☒ *New Google Calendar Event Synced to Salesforce*

```
*Event:* {{ $('Format Event Data').item.json.Subject }}
*Start:* {{ $('Format Event Data').item.json.StartDateTime }}
*End:* {{ $('Format Event Data').item.json.EndDateTime }}
*Location:* {{ $('Format Event Data').item.json.Location || 'N/A' }}
*Salesforce Event ID:* {{ $json.id }}
*Google Calendar Link:* {{ $('Format Event Data').item.json.GoogleCalendarEventLink }}
```

4. Test the Node (Optional)

- Click "Execute Node" to test
- Check your Slack channel for the message

5. Save the Node

Step 3.5: Add Error Trigger Node

1. Add Error Trigger

- Click the "+" button in an empty area of the canvas (not connected to main flow)
- Search for "Error Trigger"
- Select "Error Trigger"

2. Configure Error Trigger

- **Node Name:** `Error Trigger`
- **Continue on Fail:** Check this box (allows workflow to continue processing other events)
- This node automatically catches errors from any node in the workflow

3. Position the Node

- Drag it to a separate area below the main flow (for visual clarity)

4. Save the Node

Step 3.6: Add Send Error to Slack Node

1. Add Another Slack Node

- Click the "+" button after the Error Trigger node
- Search for "Slack"
- Select "Slack"

2. Configure the Error Slack Node

- **Node Name:** `Send Error to Slack`
- **Credential:** Select "Slack API"
- **Resource:** Select "Message"
- **Operation:** Select "Post"

3. Configure Error Message

- **Channel:** Enter channel name (e.g., `#n8n-alerts`) or use expression: `{{ $env.SLACK_CHANNEL || '#n8n-alerts' }}`
- **Text:** Click the expression icon (fx) and enter:

✕ `*Google Calendar to Salesforce Event Error*`

```
*Error:* {{ $json.error.message }}
*Node:* {{ $json.node.name }}
*Workflow:* {{ $json.workflow.name }}
*Execution ID:* {{ $json.execution.id }}

*Event Data:*
```json
{{ JSON.stringify($json.executionData, null, 2) }}
```



#### 4. **\*\*Save the Node\*\***

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### ## Part 4: Connect the Nodes

#### ### Step 4.1: Connect Main Flow

##### 1. **\*\*Connect Google Calendar Trigger → Format Event Data\*\***

- Hover over the Google Calendar Trigger node
- Click and drag from the output dot (right side) to the Format Event Data node

##### 2. **\*\*Connect Format Event Data → Create Salesforce Event\*\***

- Drag from Format Event Data output to Create Salesforce Event input

##### 3. **\*\*Connect Create Salesforce Event → Slack Alert\*\***

- Drag from Create Salesforce Event output to Slack Alert input

#### ### Step 4.2: Connect Error Flow

##### 1. **\*\*Connect Error Trigger → Send Error to Slack\*\***

- Drag from Error Trigger output to Send Error to Slack input

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### ## Part 5: Final Configuration & Testing

#### ### Step 5.1: Save the Workflow

##### 1. **\*\*Save All Changes\*\***

- Click "Save" button (top right)
- Or press Ctrl+S / Cmd+S

#### ### Step 5.2: Test the Workflow

##### 1. **\*\*Manual Test Execution\*\***

- Click "Execute Workflow" button (top right)
- This will run the workflow with sample/test data
- Review each node's output to ensure data flows correctly

##### 2. **\*\*Test with Real Google Calendar Event\*\***

- Create a test event in your Google Calendar
- Wait a few seconds
- Check n8n execution logs to see if the workflow triggered
- Verify the event was created in Salesforce
- Check Slack for notifications

#### ### Step 5.3: Activate the Workflow

##### 1. **\*\*Activate the Workflow\*\***

- Toggle the "Active" switch at the top of the workflow (or in workflow settings)

- The workflow is now live and will trigger automatically when Google Calendar events are created

## 2. **\*\*Verify Activation\*\***

- The workflow should show as "Active" in the workflows list
- The Google Calendar Trigger should show as listening for events

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## ## Part 6: Monitoring & Troubleshooting

### ### Step 6.1: Monitor Executions

#### 1. **\*\*View Execution History\*\***

- Click on "Executions" in the left sidebar
- Filter by your workflow name
- Click on any execution to see detailed logs

#### 2. **\*\*Check Node Outputs\*\***

- In execution view, click on each node to see:
  - Input data
  - Output data
  - Any errors

### ### Step 6.2: Common Issues & Solutions

#### **\*\*Issue: Workflow not triggering\*\***

- ☒ Check if workflow is "Active"
- ☒ Verify Google Calendar OAuth2 credentials are valid
- ☒ Check Google Calendar webhook is properly configured
- ☒ Verify the calendar you're monitoring has events

#### **\*\*Issue: Salesforce event not created\*\***

- ☒ Verify Salesforce OAuth2 credentials
- ☒ Check field API names are correct (especially custom fields)
- ☒ Verify you have permission to create Events in Salesforce
- ☒ Check Salesforce API limits haven't been exceeded

#### **\*\*Issue: Slack notifications not sending\*\***

- ☒ Verify Slack API credentials
- ☒ Check channel name is correct (include # for public channels)
- ☒ Verify bot has permission to post in the channel
- ☒ Check Slack API rate limits

#### **\*\*Issue: Date/time format errors\*\***

- ☒ Verify Google Calendar is sending dates in ISO 8601 format
- ☒ Check timezone handling in the Format Event Data node
- ☒ Ensure Salesforce is receiving dates in correct format

#### **\*\*Issue: Custom fields not saving\*\***

- ☒ Verify custom fields exist in Salesforce
- ☒ Check field API names match exactly (case-sensitive)
- ☒ Verify field-level security allows updates
- ☒ Ensure fields are on the Event page layout

```

Part 7: Optional Enhancements

Step 7.1: Add Event Filtering

If you only want to sync certain events:

1. Add IF Node
 - Add an "IF" node between Format Event Data and Create Salesforce Event
 - Configure condition (e.g., only sync events with specific keywords)

2. Example Filter
 - Condition: `{{ $json.Subject }}` contains "Meeting"
 - Only events with "Meeting" in the subject will proceed

Step 7.2: Add Multiple Calendar Support

1. Modify Google Calendar Trigger
 - Add multiple trigger nodes for different calendars
 - Or use a loop to process multiple calendars

Step 7.3: Add Attendee Management

1. Add Code Node After Format Event Data
 - Look up Contacts/Leads by email from attendees array
 - Create EventRelation records in Salesforce

Part 8: Workflow Summary

Your complete workflow should look like this:
```

Google Calendar Trigger ↓ Format Event Data (Code) ↓ Create Salesforce Event ↓ Slack Alert

Error Trigger (separate branch) ↓ Send Error to Slack

```

Quick Reference: Node Configuration Summary

| Node | Credential | Key Settings |
|-----|-----|-----|
| Google Calendar Trigger | Google Calendar OAuth2 | Event: "Event Created" |
| Format Event Data | None | JavaScript code for transformation |
| Create Salesforce Event | Salesforce OAuth2 | Resource: Event, Operation: Create |
| Slack Alert | Slack API | Resource: Message, Operation: Post |
```

```
| Error Trigger | None | Continue on Fail: Enabled |
| Send Error to Slack | Slack API | Resource: Message, Operation: Post |
```

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## ## Next Steps

1. ☒ Test the workflow thoroughly
2. ☒ Monitor first few real executions
3. ☒ Adjust field mappings if needed
4. ☒ Customize Slack notification format
5. ☒ Consider adding filtering or other enhancements

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## ## Support Resources

- **n8n Documentation**: <https://docs.n8n.io/>
- **Google Calendar API**: <https://developers.google.com/calendar/api>
- **Salesforce REST API**: [https://developer.salesforce.com/docs/atlas.en-us.api\\_rest.meta/api\\_rest/](https://developer.salesforce.com/docs/atlas.en-us.api_rest.meta/api_rest/)
- **Slack API**: <https://api.slack.com/>

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**\*\*Congratulations!\*\*** Your Google Calendar to Salesforce Event workflow is now set up and running! 🎉